

1. Agent Playtest 0020

Field	Value
Date	2026-10-02 04:55 – 05:20 local
Commit (local state actually running)	1f6a99f “Add ./docs/bug_reports and ./docs/skill/bugreport.md” + uncommitted working tree, per file: client/src/gameplay/player_character.rs (the fix : local_rollback registration for CharacterControllerState) · client/src/lifecycle/networking.rs + server/src/networking.rs (TLS provider switched ring → aws-lc-rs, caught during verification) · docs/agents/bug_reports/bug_0004.typ (status → Fixed, root cause corrected) · AGENTS.md (movement gap corrected)
Agent	opcode session, GLM-5.3-Flash
Clients	2x target/debug/client --mcp --no-render (dev-tools feature); client B on fleet ports :15712/:15713
Server	target/release/server — fresh
Level	levels/minimal.level.ron
Transports	game: UDP/netcode :6000 · token endpoint: HTTPS :6001 (aws-lc-rs provider) · client QA: BRP :15702 + MCP :15710 · server QA: BRP :15701

2. Purpose

Fix the last technical crack, docs/agents/bug_reports/bug_0004 (status Open at session start): ahoy’s KCC-internal state (CharacterControllerState — coyote/jump-buffer stopwatches, grounded-hit data — plus AccumulatedInput) was not registered for lightyear’s rollback, so server corrections rewound Position/Velocity while controller-internal decision state kept post-correction values → ghost jumps / swallowed buffered jumps. The working assumption had been that this **needs ahoy-side exposure and can’t be fixed from this repo**.

3. Root cause re-examination (the premise was wrong)

Direct read of bevy_ahoy 0.2.0’s source: CharacterControllerState derives Component + Clone (+ Reflect, PartialEq) with mutable mutability — exactly the requirement of lightyear 0.30’s built-in **local rollback** API, PredictionBuilderExt::local_rollback() (“enable local rollback for a component ... not handled by Replicon’s prediction marker writes”, bounds: Component<Mutability = Mutable> + Clone). No ahoy-side change, no fork, no snapshot plumbing needed. The old assessment was written against an assumed need for serializable/exposed internals; local_rollback requires none of that.

4. The fix

One registration in PlayerCharacterPlugin::build:

```
app.component::<CharacterControllerState>().local_rollback();
```

- Makes lightyear maintain PredictionHistory<CharacterControllerState> on predicted entities (snapshotted in FixedPostUpdate’s UpdateHistory, restored on rewind) **without replicating the state to anyone**.
- Ordering requirement: local_rollback registers rollback metadata only if PredictionRegistry already exists (created by PredictionPlugin inside ClientPlugins) — satisfied because PlayerCharacterPlugin is added later in main.rs’s tuple. Documented in-code.
- AccumulatedInput deliberately **not** registered: it is cleared and re-filled from the replayed input stream every tick, so it re-derives correctly during rollback — snapshotting it would only store post-consumption zeros.
- Scope: client-only by construction (local_rollback is a no-op wherever PredictionRegistry is absent, i.e. on the authoritative server).

5. Verification

Two --no-render clients, fresh server: A connects → selects level → plays; B joins (its join-burst triggers the documented acute correction window on A — the exact condition the bug needs).

Check	Observed
History registration	A's own player entity carries PredictionHistory<bevy_ahoy::CharacterControllerState> alongside the four physics histories (Position/Rotation/LinearVelocity/AngularVelocity) — present only on the local predicted player, per world.list_components
Movement sanity through corrections	A moves via game/input movement after B's join-burst corrections: position updates correctly ((0, 0.92, -8.67) after 40 ticks), grounded: true, no desync symptoms
Panics/errors	0 across both clients and the server

6. Findings

F1 — bug_0004 is fixed, and the fix is one line + a comment. The “needs ahoy-side exposure” assessment in the report and AGENTS.md was wrong: ahoy 0.2 already derived Component + Clone on CharacterControllerState, which is all local_rollback() requires. Lesson recorded in the bug report and AGENTS.md: before declaring an upstream-exposure requirement, check the **derives** against the target API's actual trait bounds.

F2 — Provider mismatch caught during verification. The client's rustls resolved with the **aws-lc-rs** crypto provider (via rmcp's feature set) while the token-endpoint code written last turn used rustls::crypto::ring::default_provider() — a compile-time catch (could not find ring in crypto). Both binaries switched to rustls::crypto::aws_lc_rs::default_provider() (already in the graph via rmcp; one provider everywhere, so client and server negotiate identically). Server re-verified over the full HTTPS token flow after the switch (token 1 issued, connect → InGame, zero failures).

F3 — Behavioral verification limit, stated honestly. The **structural** fix (history registration on the predicted player) is live-verified; the **behavioral** symptom (ghost jump vs swallowed jump across a correction) is not deterministically observable through the harness — it requires catching a jump input inside the narrow mispredicted window. The registration is the mechanism lightyear documents for exactly this state class, the history is confirmed present and per-tick updated, and movement through the correction burst is clean. A targeted ghost-jump repro (scripted jump input timed against forced corrections) would be the follow-up if symptoms are ever reported in practice.

7. Next steps

1. **Spawn-point separation** (design item, playtest 0018) — joiners still stack on an idle predecessor.
2. **Production netcode upgrades** (explicitly deferred): CA-signed cert on a real backend; asymmetric first-connection exchange if first-use MITM safety is ever required (TOFU trusts its first observer by definition).
3. **KCC rollback behavioral repro** (F3) — only if ghost-jump symptoms are ever reported.
4. Player names are generated handles (“Adjective Noun” + collision suffix); real name sourcing remains an open design item when identity matters beyond uniqueness.

8. Conclusion

The last technical crack is closed: ahoy's KCC decision state participates in rollback via lightyear's built-in local-rollback registration, live-verified on the predicted player, with the TLS provider mismatch caught and fixed during the same pass. The bug ledger now shows bug_0001–bug_0004 all Fixed, with the remaining items being explicit design decisions (spawn separation, name sourcing, production cert/keychain posture) rather than defects. AGENTS.md's movement gap bullet is corrected to match.